$n = 10110$ $\boxed{20}$

$n-1 = 10011$ $\boxed{19}$

$n \& (n-1) = 10000$ $16+2+16$

$n = 10000$

$n-1 = 01111$

$n \& (n-1) = 00000$

$(n-1) \rightarrow$ Flip the rightmost set bit to 0 & all the bits to the right of it to 1

Hence, $[n \& (n-1)] \longrightarrow$ The rightmost set bit of n goes away

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10110

RS $\rightarrow$ 1011

RS $\rightarrow$ 101

RS $\rightarrow$ 10

RS $\rightarrow$ 1

RS $\rightarrow$ 0

cnt: 0, 1, 2, 3, 4

TC = O(No. of binary digits)

$n = 10110011$

$n-1 = 10110010$

10110000

$n-1 = 10100111$

10100000

10000000

01111111

00000000

TC = O(no. of bits)

$[1, 10, 1, 1, 2, 2, 2, 4, 4, 4]$

0001

0010

0001

0010

0010

0010

0100

0100

0100

5

3

5+3 5x

5	3	4	3
1	0	1	0

3+5 $1010 = 10.$

TC = O(n)

AS = O(1)